

THIS PAGE IS FOR TEACHER USE ONLY:

Student Name:



St George Girls High School

Mathematics Advanced

2021 Year 11 Assessment Task 3

General Instructions

- Working Time – 50 minutes.
- No reading time.
- Write using black pen.
- Calculators approved by NESA may be used.
- You can use your reference sheet.
- To answer the questions
 - Answer each question on a separate A4 sheet of paper, which includes your full name, teacher's name and question number.
 - Show relevant mathematical reasoning and/or calculations.
 - Marks may not be awarded for incomplete or poorly presented solutions.

Total marks: 38

- Attempt Questions 1 – 8

Q1	/5
Q2	/5
Q3	/5
Q4	/5
Q5	/5
Q6	/5
Q7	/4
Q8	/4
Total	/38
	%

Question 1 (5 marks)

a) Differentiate with respect to x :

i) $y = 8x + \frac{1}{x^2}$. 1

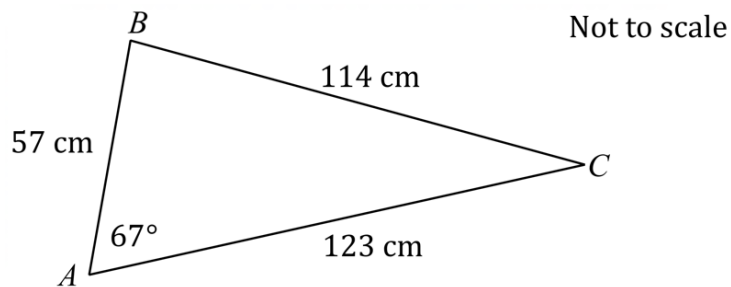
ii) $y = \frac{4}{\sqrt{3x-1}}$. 2

b) Simplify the following expression: $\frac{1 - \cos^2 \theta}{\tan \theta}$. 2

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Question 2 (5 marks) – Start your answer on a new page.

a)

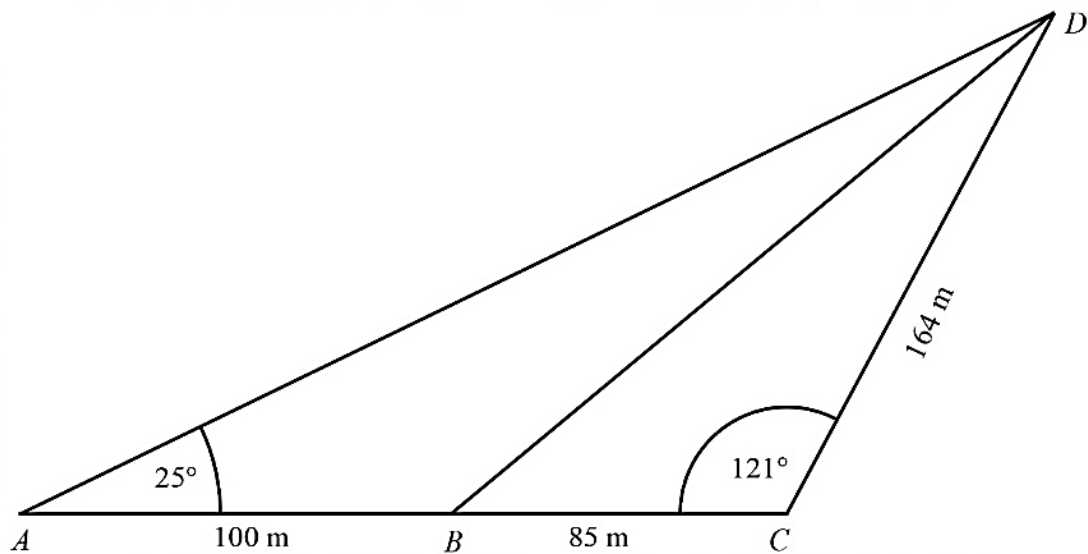


What is the area of triangle ABC to the nearest square centimetre?

1

b)

The diagram below shows four cafés in a park, labelled A , B , C and D . There are straight paths joining the cafés.



i) Show by calculations that the distance BD is 220m (to the nearest metre).

1

ii) Find the size of $\angle ADB$, correct to the nearest degree.

1

iii) Lisa walks from A to B to D . How many extra metres does she walk compared to if she had walked directly from A to D ? Answer to the nearest metre.

2

Continued on next page

Question 3 (5 marks) – Start your answer on a new page.

- a) Differentiate $f(x) = 4x^2 - 5x$, using the method of first principles. 3
- b) If $y = (2x + 1)(3x + 4)^3$, find $\frac{dy}{dx}$. 2

Question 4 (5 marks) – Start your answer on a new page.

A function is defined by $f(x) = \frac{2x - 1}{3x + 2}$.

- i) Show that $f'(x) = \frac{7}{(3x + 2)^2}$. 2
- ii) Find $f'(-1)$. 1
- iii) Find the equation of the tangent at the point where $x = -1$. 2

Continued on next page

Question 5 (5 marks) – Start your answer on a new page.

- a) A particle is moving along the x -axis.
Its displacement x centimetres at time t seconds is given by:

$$x = 4 + 12t^2 - t^3 \text{ for } t \geq 0.$$

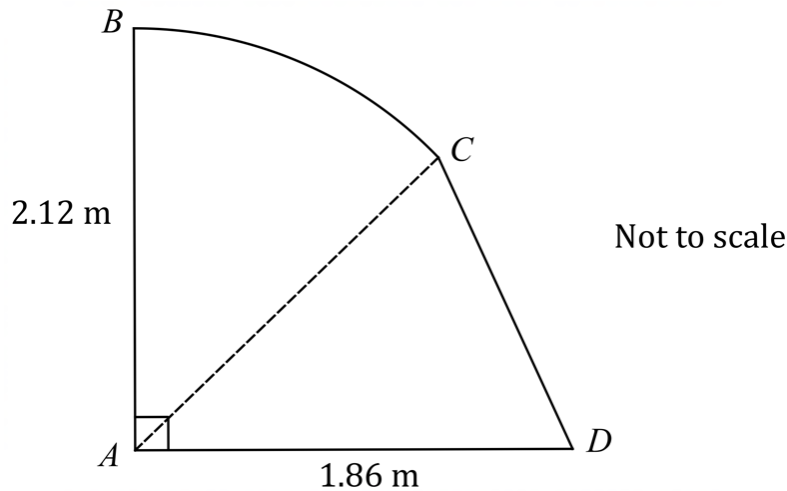
- | | | |
|-----|---|---|
| i) | What is the initial displacement of the particle? | 1 |
| ii) | At what time(s) is the particle at rest? | 2 |

- b) Solve the equation $2 \cos \alpha - \sqrt{3} = 0$ for $0 \leq \alpha \leq 2\pi$. 2

Continued on next page

Question 6 (5 marks) – Start your answer on a new page.

a)



A plane shape $ABCD$ is shown above.

The straight line AB is vertical and has length 2.12 metres.

The straight line AD is horizontal and has length 1.86 metres.

The curve BC is an arc of a circle with centre A , and CD is a straight line.

The size of $\angle BAC$ is 0.65 radians.

i) Find the length of the arc BC . Answer correct to two decimal places. 1

ii) What is the area of the sector BAC ? 1
Answer correct to two decimal places.

iii) Determine the size of $\angle CAD$, in radians, correct to two decimal places. 1

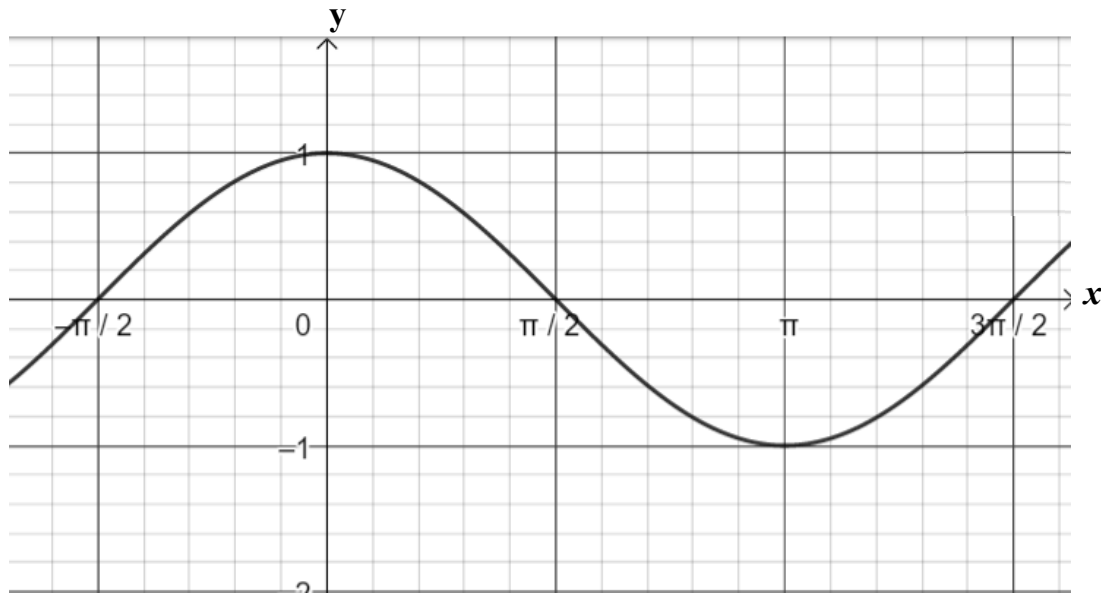
b) If $\sin \theta = \frac{7}{24}$ and $\cos \theta < 0$, find the exact value of $\sec \theta$. 2

Continued on next page

Question 7 (4 marks) – Start your answer on a new page.

- a) Write an equation for the graph below.

1



- b) A block of ice of mass 8 kg melts according to the formula $M = 8 - \frac{t^2}{32}$, 3

where M is the mass remaining, in kilograms, after t minutes.

Find the rate at which the ice is melting when 2kg of ice has melted.

Continued on next page

Question 8 (4 marks) – Start your answer on a new page.

a) Let $f(x) = |x - 2|$.

i) Explain why this function is continuous at $x = 2$. 1

ii) Is this function differentiable at $x = 2$? Explain your answer. 1

b) The volume of water in a container after t minutes is given by the equation 2

$V(t) = t^2 + 4t + 30$ where V is measured in litres.

It is known that the average rate of change in volume over the first k minutes is 10 litres per minute. Find the value of k .

End of Assessment Task

Mathematics Advanced
Mathematics Extension 1
Mathematics Extension 2

REFERENCE SHEET

Measurement

Length

$$l = \frac{\theta}{360} \times 2\pi r$$

Area

$$A = \frac{\theta}{360} \times \pi r^2$$

$$A = \frac{h}{2}(a + b)$$

Surface area

$$A = 2\pi r^2 + 2\pi rh$$

$$A = 4\pi r^2$$

Volume

$$V = \frac{1}{3}Ah$$

$$V = \frac{4}{3}\pi r^3$$

Functions

$$x = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}$$

For $ax^3 + bx^2 + cx + d = 0$:

$$\alpha + \beta + \gamma = -\frac{b}{a}$$

$$\alpha\beta + \alpha\gamma + \beta\gamma = \frac{c}{a}$$

$$\text{and } \alpha\beta\gamma = -\frac{d}{a}$$

Relations

$$(x - h)^2 + (y - k)^2 = r^2$$

Financial Mathematics

$$A = P(1 + r)^n$$

Sequences and series

$$T_n = a + (n - 1)d$$

$$S_n = \frac{n}{2}[2a + (n - 1)d] = \frac{n}{2}(a + l)$$

$$T_n = ar^{n-1}$$

$$S_n = \frac{a(1 - r^n)}{1 - r} = \frac{a(r^n - 1)}{r - 1}, r \neq 1$$

$$S = \frac{a}{1 - r}, |r| < 1$$

Logarithmic and Exponential Functions

$$\log_a a^x = x = a^{\log_a x}$$

$$\log_a x = \frac{\log_b x}{\log_b a}$$

$$a^x = e^{x \ln a}$$

Trigonometric Functions

$$\sin A = \frac{\text{opp}}{\text{hyp}}, \quad \cos A = \frac{\text{adj}}{\text{hyp}}, \quad \tan A = \frac{\text{opp}}{\text{adj}}$$

$$A = \frac{1}{2}ab \sin C$$

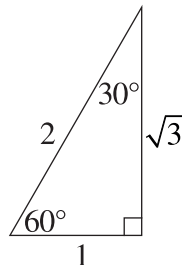
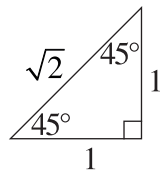
$$\frac{a}{\sin A} = \frac{b}{\sin B} = \frac{c}{\sin C}$$

$$c^2 = a^2 + b^2 - 2ab \cos C$$

$$\cos C = \frac{a^2 + b^2 - c^2}{2ab}$$

$$l = r\theta$$

$$A = \frac{1}{2}r^2\theta$$



Trigonometric identities

$$\sec A = \frac{1}{\cos A}, \quad \cos A \neq 0$$

$$\operatorname{cosec} A = \frac{1}{\sin A}, \quad \sin A \neq 0$$

$$\cot A = \frac{\cos A}{\sin A}, \quad \sin A \neq 0$$

$$\cos^2 x + \sin^2 x = 1$$

Compound angles

$$\sin(A + B) = \sin A \cos B + \cos A \sin B$$

$$\cos(A + B) = \cos A \cos B - \sin A \sin B$$

$$\tan(A + B) = \frac{\tan A + \tan B}{1 - \tan A \tan B}$$

$$\text{If } t = \tan \frac{A}{2} \text{ then } \sin A = \frac{2t}{1 + t^2}$$

$$\cos A = \frac{1 - t^2}{1 + t^2}$$

$$\tan A = \frac{2t}{1 - t^2}$$

$$\cos A \cos B = \frac{1}{2} [\cos(A - B) + \cos(A + B)]$$

$$\sin A \sin B = \frac{1}{2} [\cos(A - B) - \cos(A + B)]$$

$$\sin A \cos B = \frac{1}{2} [\sin(A + B) + \sin(A - B)]$$

$$\cos A \sin B = \frac{1}{2} [\sin(A + B) - \sin(A - B)]$$

$$\sin^2 nx = \frac{1}{2} (1 - \cos 2nx)$$

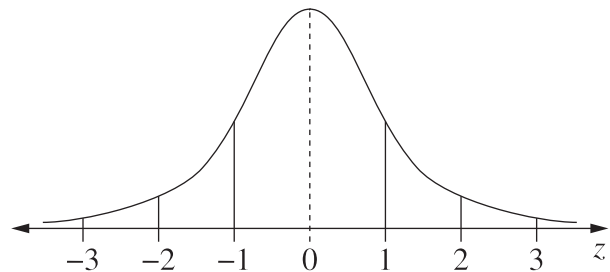
$$\cos^2 nx = \frac{1}{2} (1 + \cos 2nx)$$

Statistical Analysis

$$z = \frac{x - \mu}{\sigma}$$

An outlier is a score
less than $Q_1 - 1.5 \times IQR$
or
more than $Q_3 + 1.5 \times IQR$

Normal distribution



- approximately 68% of scores have z -scores between -1 and 1
- approximately 95% of scores have z -scores between -2 and 2
- approximately 99.7% of scores have z -scores between -3 and 3

$$E(X) = \mu$$

$$\operatorname{Var}(X) = E[(X - \mu)^2] = E(X^2) - \mu^2$$

Probability

$$P(A \cap B) = P(A)P(B)$$

$$P(A \cup B) = P(A) + P(B) - P(A \cap B)$$

$$P(A|B) = \frac{P(A \cap B)}{P(B)}, \quad P(B) \neq 0$$

Continuous random variables

$$P(X \leq r) = \int_a^r f(x) dx$$

$$P(a < X < b) = \int_a^b f(x) dx$$

Binomial distribution

$$P(X = r) = {}^nC_r p^r (1 - p)^{n-r}$$

$$X \sim \operatorname{Bin}(n, p)$$

$$\Rightarrow P(X = x)$$

$$= \binom{n}{x} p^x (1 - p)^{n-x}, \quad x = 0, 1, \dots, n$$

$$E(X) = np$$

$$\operatorname{Var}(X) = np(1 - p)$$

Differential Calculus

Function

Derivative

$$y = f(x)^n$$

$$\frac{dy}{dx} = n f'(x) [f(x)]^{n-1}$$

$$y = uv$$

$$\frac{dy}{dx} = u \frac{dv}{dx} + v \frac{du}{dx}$$

$$y = g(u) \text{ where } u = f(x)$$

$$\frac{dy}{dx} = \frac{dy}{du} \times \frac{du}{dx}$$

$$y = \frac{u}{v}$$

$$\frac{dy}{dx} = \frac{v \frac{du}{dx} - u \frac{dv}{dx}}{v^2}$$

$$y = \sin f(x)$$

$$\frac{dy}{dx} = f'(x) \cos f(x)$$

$$y = \cos f(x)$$

$$\frac{dy}{dx} = -f'(x) \sin f(x)$$

$$y = \tan f(x)$$

$$\frac{dy}{dx} = f'(x) \sec^2 f(x)$$

$$y = e^{f(x)}$$

$$\frac{dy}{dx} = f'(x) e^{f(x)}$$

$$y = \ln f(x)$$

$$\frac{dy}{dx} = \frac{f'(x)}{f(x)}$$

$$y = a^{f(x)}$$

$$\frac{dy}{dx} = (\ln a) f'(x) a^{f(x)}$$

$$y = \log_a f(x)$$

$$\frac{dy}{dx} = \frac{f'(x)}{(\ln a) f(x)}$$

$$y = \sin^{-1} f(x)$$

$$\frac{dy}{dx} = \frac{f'(x)}{\sqrt{1 - [f(x)]^2}}$$

$$y = \cos^{-1} f(x)$$

$$\frac{dy}{dx} = -\frac{f'(x)}{\sqrt{1 - [f(x)]^2}}$$

$$y = \tan^{-1} f(x)$$

$$\frac{dy}{dx} = \frac{f'(x)}{1 + [f(x)]^2}$$

Integral Calculus

$$\int f'(x) [f(x)]^n dx = \frac{1}{n+1} [f(x)]^{n+1} + c$$

where $n \neq -1$

$$\int f'(x) \sin f(x) dx = -\cos f(x) + c$$

$$\int f'(x) \cos f(x) dx = \sin f(x) + c$$

$$\int f'(x) \sec^2 f(x) dx = \tan f(x) + c$$

$$\int f'(x) e^{f(x)} dx = e^{f(x)} + c$$

$$\int \frac{f'(x)}{f(x)} dx = \ln |f(x)| + c$$

$$\int f'(x) a^{f(x)} dx = \frac{a^{f(x)}}{\ln a} + c$$

$$\int \frac{f'(x)}{\sqrt{a^2 - [f(x)]^2}} dx = \sin^{-1} \frac{f(x)}{a} + c$$

$$\int \frac{f'(x)}{a^2 + [f(x)]^2} dx = \frac{1}{a} \tan^{-1} \frac{f(x)}{a} + c$$

$$\int u \frac{dv}{dx} dx = uv - \int v \frac{du}{dx} dx$$

$$\int_a^b f(x) dx$$

$$\approx \frac{b-a}{2n} \left\{ f(a) + f(b) + 2[f(x_1) + \cdots + f(x_{n-1})] \right\}$$

where $a = x_0$ and $b = x_n$

Combinatorics

$${}_nP_r = \frac{n!}{(n-r)!}$$

$$\binom{n}{r} = {}_nC_r = \frac{n!}{r!(n-r)!}$$

$$(x+a)^n = x^n + \binom{n}{1}x^{n-1}a + \cdots + \binom{n}{r}x^{n-r}a^r + \cdots + a^n$$

Vectors

$$|\underline{u}| = |x\underline{i} + y\underline{j}| = \sqrt{x^2 + y^2}$$

$$\underline{u} \cdot \underline{v} = |\underline{u}| |\underline{v}| \cos \theta = x_1x_2 + y_1y_2,$$

$$\text{where } \underline{u} = x_1\underline{i} + y_1\underline{j}$$

$$\text{and } \underline{v} = x_2\underline{i} + y_2\underline{j}$$

$$\underline{r} = \underline{a} + \lambda \underline{b}$$

Complex Numbers

$$\begin{aligned} z = a + ib &= r(\cos \theta + i \sin \theta) \\ &= re^{i\theta} \end{aligned}$$

$$\begin{aligned} [r(\cos \theta + i \sin \theta)]^n &= r^n(\cos n\theta + i \sin n\theta) \\ &= r^n e^{in\theta} \end{aligned}$$

Mechanics

$$\frac{d^2x}{dt^2} = \frac{dv}{dt} = v \frac{dv}{dx} = \frac{d}{dx} \left(\frac{1}{2} v^2 \right)$$

$$x = a \cos(nt + \alpha) + c$$

$$x = a \sin(nt + \alpha) + c$$

$$\ddot{x} = -n^2(x - c)$$

MATHEMATICS ADVANCED – QUESTION 1

SUGGESTED SOLUTIONS	MARKS	MARKER'S COMMENTS
1 a) Differentiate wr.t.x (w.r.t.x)		
(i) $y = 8x + x^2$		
$y = 8x + x^{-2}$		No half marks
$y' = 8 - 2x^{-3}$	1	given.
		generally well done
(ii) $y = \frac{4}{\sqrt{3x-1}}$	2	
$y = 4(3x-1)^{-\frac{1}{2}}$		1 mark for this line
$y' = 4 \times -\frac{1}{2} (3x-1)^{-\frac{1}{2}-1} \times 3$		
$y' = -2(3x-1)^{-\frac{3}{2}} \times 3$		
$y' = -6(3x-1)^{-\frac{3}{2}}$		
or		
$= -\frac{6}{(\sqrt{3x-1})^3}$		1 mark answer
		many students forgot to multiply by 3 $\frac{d}{dx}(3x-1)$
(b) Simplify $\frac{1-\cos^2\theta}{\tan\theta}$	2	
$= \frac{\sin^2\theta}{\tan\theta}$ or $\frac{\sin^2\theta}{\tan\theta}$		1 mark to here
$= \frac{\sin^2\theta}{\frac{\sin\theta}{\cos\theta}} \times \frac{\cos\theta}{\cos\theta} = \sin^2\theta \div \tan\theta$		well done generally
$= \frac{\sin^2\theta \cos\theta}{\sin\theta} = \sin^2\theta \times \frac{\cos\theta}{\sin\theta}$		A number of students didn't use
$= \sin\theta \cos\theta = \sin\theta \cos\theta$		$\tan\theta = \frac{\sin\theta}{\cos\theta}$

MATHEMATICS ADVANCED – QUESTION 2

SUGGESTED SOLUTIONS	MARKS	MARKER'S COMMENTS
2a) $A = \frac{1}{2} bc \sin A$ $= \frac{1}{2} \times 123 \times 57 \times \sin 67$ $\doteq 3227 \text{ cm}^2$	1	Some students forgot to use the formula. Some did not realise the formula means 2 sides and an included angle, and so proceeded to find $\angle C$ before using formula.
2 b) i) Using $\triangle BDC$: cosine rule: $c^2 = a^2 + b^2 - 2ab \cos C$ $\therefore BD^2 = 85^2 + 164^2 - 2 \times 85 \times 164 \times \cos 121^\circ$ $\therefore BD \doteq 220 \text{ m}$	1	If students miscalculated and left another answer for BD, then $\frac{1}{2}$ was deducted.
2 b) ii) most used $\triangle ADB$ as follows $\frac{\sin \angle ADB}{100} = \frac{\sin 25^\circ}{220}$ $\therefore \angle ADB \doteq 11^\circ \text{ (nearest degree)}$	1	
There was an error in the question, and $\angle DAB$ is not equal to 25°. Therefore, the following methods were also suitable, but resulted in different values.		

MATHEMATICS ADVANCED – QUESTION 2

SUGGESTED SOLUTIONS	MARKS	MARKER'S COMMENTS
2 b) ii) continued.		
<u>Option 2</u>		
$\angle ADC = 180 - \angle DAC - \angle DCA$ $= 180 - 25 - 121$ $= 34^\circ$		
$\frac{\sin \angle BDC}{85} = \frac{\sin 121}{220}$		
$\therefore \angle BDC = 19^\circ 20'$		
$\therefore \angle ADB = \angle ADC - \angle BDC$ $= 34^\circ - 19^\circ 20'$ $= 14^\circ 40'$ $\hat{=} 15^\circ \text{ (nearest degree)}$	1	
<u>Option 3</u>		
$\frac{\sin \angle DBC}{164} = \frac{\sin 121^\circ}{220}$		
$\angle DBC = 39^\circ 43'$		
$\therefore \angle ABD = 180 - 39^\circ 43'$ $= 140^\circ 17'$		
$\therefore \angle ADB = 180 - \angle DAB - \angle ABD$ $= 180 - 25 - 140^\circ 17'$ $= 14^\circ 43'$ $\hat{=} 15^\circ$	1	
2 b) iii) – Due to the error of 25° in the diagram, this part of the question became infeasible to mark. 2 marks were awarded to all students.	2	

MATHEMATICS ADVANCED - QUESTION 3 Solutions and marking criteria

SUGGESTED SOLUTIONS	MARKS	MARKER'S COMMENTS
$a) f(x) = 4x^2 - 5x$ $f(x+h) = 4(x+h)^2 - 5(x+h)$ $\therefore f'(x) = \lim_{h \rightarrow 0} \frac{f(x+h) - f(x)}{h}$ $= \lim_{h \rightarrow 0} \frac{4(x+h)^2 - 5(x+h) - [4x^2 - 5x]}{h}$ $= \lim_{h \rightarrow 0} \frac{4(x^2 + 2xh + h^2) - 5x - 5h - 4x^2 + 5x}{h}$ $= \lim_{h \rightarrow 0} \frac{\cancel{4x^2} + 8xh + 4h^2 - \cancel{5x} - 5h - \cancel{4x^2} + \cancel{5x}}{h}$ $= \lim_{h \rightarrow 0} \frac{8xh + 4h^2 - 5h}{h}$ $= \lim_{h \rightarrow 0} \frac{\cancel{h}(8x + 4h - 5)}{\cancel{h}}$ $= \lim_{h \rightarrow 0} 8x + 4h - 5$ $= 8x - 5$	1 mark 1 mark 1	3 Note: - Many students did not have the limit sign using this principle $\therefore$ <u>-1 mark</u> or they had it appearing once or placed before the equal sign. - A handful of students did not include the vinculum to cover the whole expression $\therefore$ <u>-1/2 mark</u>. Finding the derivative from first principles needs to be reviewed by many students.

MATHEMATICS ADVANCED – QUESTION 3

SUGGESTED SOLUTIONS

MARKS

MARKER'S COMMENTS

b)

$$y = (2x+1)(3x+4)^3$$

$$u = 2x+1 \quad v = (3x+4)^3$$

$$u' = 2 \quad v' = 3(3x+4)^2 \times 3 \\ = 9(3x+4)^2$$

$$\frac{dy}{dx} = vu' + uv'$$

$$= (3x+4)^3 \times 2 + (2x+1) \times 3(3x+4)^2 \times 3$$

$$= 2(3x+4)^3 + (2x+1) \times 9(3x+4)^2 \quad \text{--- 2 marks}$$

①

①

$$= (3x+4)^2 [2(3x+4) + 9(2x+1)]$$

$$= (3x+4)^2 (6x+8+18x+9)$$

$$= (3x+4)^2 (24x+17) \quad \#$$

NOTE:

Many students successfully got to the last line by factorising and simplifying correctly however there were a handful of students who need a lot of practising this skill. Only the first two lines of working gave you the full marks though.

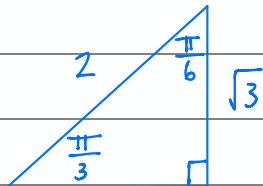
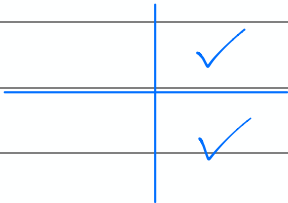
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MATHEMATICS ADVANCED – QUESTION 4 Year 11

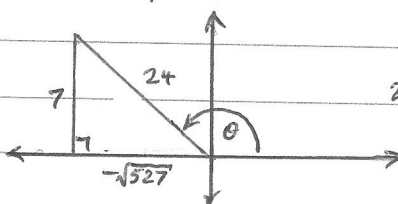
SUGGESTED SOLUTIONS	MARKS	MARKER'S COMMENTS
i) $f(x) = \frac{2x-1}{3x+2}$		Some students
$f'(x)$ - Use quotient rule.		did not
Let $u = 2x-1$ $v = 3x+2$		have the
$\frac{du}{dx} = 2$ $\frac{dv}{dx} = 3$		correct
$f'(x) = \frac{v \frac{du}{dx} - u \frac{dv}{dx}}{v^2}$		quotient rule formula.
		It is on
		the reference
		sheet. Others
$= \frac{2(3x+2) - 3(2x-1)}{(3x+2)^2}$	1mk	made mistakes
$= \frac{6x+4 - 6x+3}{(3x+2)^2}$	Correct application in the of quotient rule 1mk	
$= \frac{7}{(3x+2)^2}$	for expansion and simplification	
As required		
ii) $f'(-1) = \frac{7}{(3(-1)+2)^2}$		Generally
$= \frac{7}{(-3+2)^2}$		well attempted
$= \frac{7}{(-1)^2}$		
$= 7$	1mk	Many
iii) When $x = -1$, $f(-1) = \frac{2(-1)-1}{3(-1)+2}$		computational
$= \frac{-3}{-1} = 3$	1mk	errors were
		made by
		students,
$y - 3 = 7(x - -1)$		otherwise well
$y - 3 = 7x + 7$		done
$y = 7x + 10$		
or $y - 7x - 10 = 0$	1mk	

MATHEMATICS ADVANCED – QUESTION 5

5

SUGGESTED SOLUTIONS	MARKS	MARKER'S COMMENTS
(a) $x = 4 + 12t^2 - t^3$		
(i) when $t = 0$, $x = 4 + 12(0)^2 - (0)^3$ $\therefore x = 4 \text{ cm}$ $\therefore$ Initial displacement is 4cm.	1	(i) "Initial" means when $t = 0$. Students should also show the substitution
(ii) $v = \frac{dx}{dt} = 24t - 3t^2$ particle is at rest when $v = 0$. $24t - 3t^2 = 0$ $3t(8 - t) = 0$ $\therefore t = 0, t = 8$ $\therefore$ Particle is at rest at 0 seconds and 8 seconds.	1	(ii) "At rest" means when the particle is stationary $\Rightarrow v = 0$. Be careful to use the correct notation: $\frac{dy}{dx}$ $v = \frac{dx}{dt} = \dot{x}$
(b) $2\cos\alpha - \sqrt{3} = 0$ for $0 \leq \alpha \leq 2\pi$ $2\cos\alpha = \sqrt{3}$ $\cos\alpha = \frac{\sqrt{3}}{2}$	1	- If the domain is $0 \leq \alpha \leq 2\pi$, the angles are measured in RADIANS.
 related angle = $\frac{\pi}{6}$		- Some students forgot about the solution in the 4th quadrant.
 Solutions exist in 1st and 4th quadrants. $\alpha = \frac{\pi}{6}, 2\pi - \frac{\pi}{6}$ $\therefore \alpha = \frac{\pi}{6}, \frac{11\pi}{6}$	1	- Be careful with the notation: Not $\cos\alpha = \frac{\pi}{6}$ but $\alpha = \frac{\pi}{6}$

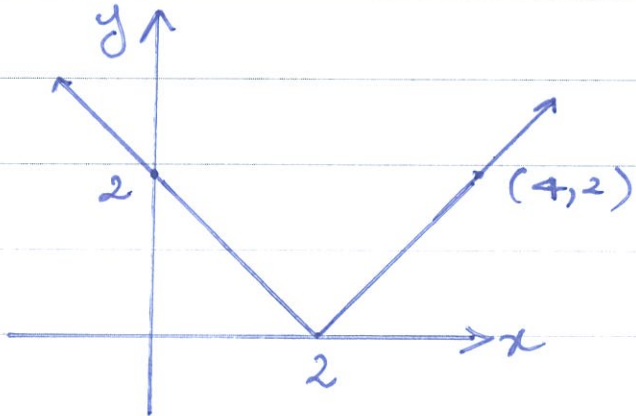
MATHEMATICS ADVANCED – QUESTION 6

SUGGESTED SOLUTIONS	MARKS	MARKER'S COMMENTS
a) (i) $l = r\theta$ $r = 2.12$, $\theta = 0.65$ $= 2.12 \times 0.65$ $= 1.378$ $= 1.38 \text{ correct to 2 dec. pl.}$	1	Some students were confused with the use of radians
(ii) $\text{Area} = \frac{1}{2} r^2 \theta$ $= \frac{1}{2} \times 2.12^2 \times 0.65$ $= 1.46068$ $= 1.46 \text{ correct to 2 dec. pl.}$	1	Same as for (i)
(iii) $\angle BAD = \frac{\pi}{2}$ radians $\angle CAD = \angle BAD - \angle BAC$ $= \frac{\pi}{2} - 0.65$ $= 0.9207963\dots$ $= 0.92 \text{ correct to 2 dec. pl.}$	1	
b) $\sin \theta = \frac{7}{24}$ $\cos \theta < 0$ second quadrant  $\cos \theta = -\frac{\sqrt{527}}{24}$ $\sec \theta = -\frac{24}{\sqrt{527}}$	2	1 mark for finding the length of the adjacent side and setting up the secant ratio 1 mark for making the answer negative

MATHEMATICS – QUESTION 7

SUGGESTED SOLUTIONS	MARKS	MARKER'S COMMENTS
a) $y = \cos x$	①	Correct Answer
b) $M = 8 - \frac{t^2}{32}$		$\frac{1}{2}$ mark if
$6 = 8 - \frac{t^2}{32}$		there was
		no $y = \cos x$
		(left off $y =$)
		and $\frac{1}{2}$ mark
		if it was not
		<u>cos</u> (ie <u>cos</u>)
NOTE: M is the		
mass "REMAINING"		
$t^2 = 64$		
$t = 8$ ($t \geq 0$ as t is time)	①	Realising $M = 6$ kg
$\therefore$ It takes 8 min		
for the 8 kg to melt	①	Finding the time
to 6 kg.		(get used to
		justifying that $t \geq 0$
$M = 8 - \frac{1}{32}t^2$		as t is time) no
$\frac{dM}{dt} = -\frac{1}{16}t$		marks deducted this
	①/2	time.
$\frac{dM}{dt} = -\frac{1}{16}t$		
$= -\frac{1}{16} \times 8$		
$= -\frac{1}{2} \text{ kg/min}$	①/2	Correct Answer
		(rate must be
$\therefore$ The rate at which the ice is		evident with
melting when 2 kg of ice has melted		units kg/min)
is 0.5 kg/min .		Also if you
		say "melting" at
		0.5 kg/min , the
		word "melting"
		means the
		negative sign.
		(no marks deducted
		for this as long
		at $\frac{dM}{dt} = -0.5 \text{ kg/min}$

MATHEMATICS ADVANCED – QUESTION 8 (4 marks)

SUGGESTED SOLUTIONS	MARKS	MARKER'S COMMENTS
(a)(i) $f(x) = x-2$ is continuous because it can be drawn without lifting your pen from the paper.	①	for correct explanation.
<u>OR</u> $f(x) = x-2$ is continuous because it is made up of two lines $y = x-2$ and $y = -x+2$ joining at $x=2$.		① for correct explanation
<u>OR</u> 		
The graph as shown does not have any breaks or gaps.	①	for correct explanation with correct graph drawn

MATHEMATICS ADVANCED – QUESTION 8

SUGGESTED SOLUTIONS	MARKS	MARKER'S COMMENTS
<u>OR</u> <u>using</u> Formal definition: $f(x) = x-2 = \begin{cases} +(x-2) & \text{for } x-2 \geq 0 \\ & x \geq 2 \\ -(x-2) & \text{for } x-2 < 0 \\ & x < 2 \end{cases}$ $\lim_{x \rightarrow 2^+} f(x) = 2-2 = 0$ $\lim_{x \rightarrow 2^-} f(x) = -2+2 = 0$ Since, $\lim_{x \rightarrow 2^+} f(x) = \lim_{x \rightarrow 2^-} f(x) = 0$, $f(x) = x-2$ is continuous at $x=2$.		
(ii) No, $f(x) = x-2$ is not differentiable at $x=2$, because there is a sharp corner at that point.	①	for correct explanation.
<u>OR</u> Not differentiable at $x=2$, since the function is made up of 2 lines with different gradients	①	for correct explanation.

MATHEMATICS ADVANCED – QUESTION 8

SUGGESTED SOLUTIONS

MARKS

MARKER'S COMMENTS

or Using formal definition:

$$f(x) = \begin{cases} x-2 & \text{for } x \geq 2 \\ -x+2 & \text{for } x < 2 \end{cases}$$

$$\therefore f'(x) = \begin{cases} 1 & \text{for } x \geq 2 \\ -1 & \text{for } x < 2 \end{cases}$$

$$\lim_{x \rightarrow 2^-} f'(x) = -1$$

① provides correct explanation.

$$\lim_{x \rightarrow 2^+} f'(x) = 1$$

Since $\lim_{x \rightarrow 2^-} f'(x) \neq \lim_{x \rightarrow 2^+} f'(x)$, then

$f(x) = |x-2|$ is not differentiable at $x=2$.

MATHEMATICS ADVANCED – QUESTION 8

SUGGESTED SOLUTIONS	MARKS	MARKER'S COMMENTS
(b) when $t=0$ (initially):		
$v(0) = (0)^2 + 4(0) + 30 = 30$ litres		
when $t=k$:		
$v(k) = k^2 + 4k + 30$		
<u>Average rate of</u>		
Change : $\frac{v(k) - v(0)}{k - 0} = 10$ L/min		
$\frac{k^2 + 4k + 30 - 30}{k} = 10$		(2) marks provides correct solution
$k^2 + 4k = 10k$		
$k^2 - 6k = 0$		
$k(k-6) = 0$		(1) mark for some worthwhile progress
$k=0$ or $k=6$		
$\therefore k=6$ minutes (since $k > 0$)		
• students need to be mindful of their algebra skills.		
<u>Remember</u> : Average rate of change		
$= \frac{\text{change in } y\text{-value}}{\text{change in time}}$		